Yasaman Kargar Gaz Kooh

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BIOGRAPHY

I am a **forth-year Ph.D. candidate** in the **Institute of Material Science and Engineering (IMSE)** at **Washington University in St. Louis**, advised by **Prof. Nathaniel Huebsch**. My research focuses on how mechanical cues influence heart development and disease progression. My current areas of interest include Cardiac Tissue Engineering, Cellular/Molecular Biology, Mechanobiology, and Stem cell Engineering.

EDUCATION

Washington University in St. Louis, St. Louis, MO, USA. **Sept. 2022 – Expected 2029**
Ph.D. Candidate in Materials Science and Engineering
Advisor: Prof. Nathaniel Huebsch

Washington University in St. Louis, St. Louis, MO, USA. **Sept. 2022 – Dec. 2024**
M.Sc. in Materials Science and Engineering
Advisor: Prof. Nathaniel Huebsch

Sharif University of Technology, Tehran, Iran. **Aug. 2017 – Sep. 2022**
B.Sc. in Materials Science and Engineering
Advisor: Prof. Abdolreza (Arash) Simchi

HONORS AND AWARDS

- **795th place out of 200,000 (top 0.5%)** competitors in National University Entrance Exam, Summer 2017 Admission to Sharif University of Technology B.Sc. program and full scholarship
- **Third Place** in the research competition on the **Concentration of Low-Grade Hematite Soil**, Sharif University of Technology, Tehran, Iran, Summer 2019
- **Second Place**, Best Poster Award, Engineering in Cardiovascular Medicine, Ann Arbor, Michigan, USA, Summer 2026

Research Funding

- **ICTS Just-In-Time (JIT) Core Usage Funding Award**, Institute of Clinical and Translational Sciences, Washington University in St. Louis, 2026–2027, "*Identifying a Transcriptomic Signature of Desmin Related Cardiomyopathy Using iPSC Cardiomyocytes*"
  Role: Co-Investigator (PI: Nathaniel Huebsch) | JIT1412

PUBLICATIONS

('*' indicates equal contribution)

Preprint

{a1} **Yasaman Kargar Gaz Kooh**, Yuwen Sun, Ganesh Malayath, Leah Hayem, Bahareh Bahmani, Nongmaithem Debeni Devi, Huanzhu Jiang, Feng Long, Shaon Cresci, Guy Genin, Nathaniel Huebsch. "*Tunable Magnetorheological Elastomer Substrates for Studying Mechano-Genetic Interactions in Hypertrophic Cardiomyopathy*". doi: 10.64898/2026.10.24.761361, in Review *Bioactive Materials (IF:23.6)*

{a2} **Yasaman Kargar Gaz Kooh**, Bahareh Bahmani, Chen Zhao, Ganesh Malayath, Leah Hayem, Hanxun Jin, Ghiska Ramahdita, Huanzhu Jiang, Javier Santiago Perez, Hsin-Yi Chou, Xiucui Ma, Guy Genin, David Rawnsley, Abhinav Diwan, Nathaniel Huebsch. "*A Micro-Engineered Heart Tissue Model of Desmin-related Cardiomyopathy Caused by Mutant B Crystalin*". doi: 10.1101/2025.10.02.680018, Accepted in *Science Advances (IF:13.9)*

{a3} Mohammadjafar Hashemi, Nongmaithem Debeni Devi, **Yasaman Kargar Gaz Kooh**, Charlotte Chen, Bahareh Bahmani, Ganesh Malayath, Julia Victor, Nathaniel Huebsch. "*Xeno-Free Peptide-Functionalized Hydrogels Support hiPSC Encapsulation and In Situ Differentiation into Structurally Mature Cardiomyocytes*". doi: 10.64898/2026.07.08.737331, in Review *Advanced Materials (IF:29)*

Journal

{b1} David R Rawnsley*, Moydul Islam*, Chen Zhao*, Xumin Guan*, **Yasaman Kargar Gaz Kooh**, Adelita Mendoza, Honora Navid, Minu Kumari, Phalgun Pandi, John T Murphy, Jess Nigro, Attila Kovacs, Lina Greenberg, Kartik Mani, Michael Greenberg, Nathaniel Huebsch, Xiucui Ma, Abhinav Diwan. "*Mitophagy Facilitates Cytosolic Proteostasis to Preserve Cardiac Function*". ***Circulation Research (IF:18)***. doi: 10.1161/CIRCRESAHA.126.328328

{b2} Huanzhu Jiang, Ganesh Malayath, Nongmaithem Debeni Devi, Hsin-Yi Chou, **Yasaman Kargar Gaz Kooh**, Jingxuan Guo, Ghiska Ramahdita, Jenna Nguyen, Riya Bhakta, Lavanya Aryan, Sharon Cresci, Nathaniel Huebsch . "*Biomechanical Stress Unmasks a Fibroblast-Dependent Hypercontractile-Disarray Phenotype in MYBPC3 Truncation HCM* ". ***APL Bioengineering***. doi: 10.1063/5.0336866

{b3} **Yasaman Kargar Gaz Kooh**, Nathaniel Huebsch. "*Bridging High Resolution Sub-cellular Imaging with Physiologically Relevant Engineered Tissues*". ***Front. Mech. Eng.*** doi: 10.3389/fmech.2024.1481933

{b4} David Schuftan, **Yasaman Kargar Gaz Kooh**, Jingxuan Guo, Yuwen Sun, Lavanya Aryan, Bryce Stottlemire, Cory Berkland, Guy M Genin, Nathaniel Huebsch. "*Dynamic Control of Contractile Resistance to iPSC-Derived Micro-heart Muscle Arrays*". ***JBMR part A***. doi: 10.1002/jbm.a.37642

Abstract

{c1} **Yasaman Kargar Gaz Kooh**, Bahareh Bahmani, Chen Zhao, Ganesh Malayath, Anastasia Khokhlova, Leah Hayem, Hanxun Jin, Ghiska Ramahdita, Huanzhu Jiang, Javier Santiago Perez, Hsin-Yi Chou, Jonathan Silva, Xiucui Ma, Guy M. Genin, David Rawnsley, Abhinav Diwan, Nathaniel Huebsch. "*Low-dose JAK1 inhibitor partially rescues contractility in a human cardiac micro-tissue model of Desmin-related Cardiomyopathy caused by CRYAB mutation*". ***Circulation Research***. BCVS, 2026

{c2} **Yasaman Kargar Gaz Kooh**, Bahareh Bahmani, Chen Zhao, Ganesh Malayath, Leah Hayem, Ghiska Ramahdita, Huanzhu Jiang, Javier Santiago Perez, Hsin-Yi Chou, Yuxuan Huang, Xiucui Ma, Guy Genin, David Rawnsley, Abhinav Diwan, Nathaniel Huebsch. "*A Micro-Engineered Heart Tissue Model of Desmin-related Cardiomyopathy Caused by Mutant B Crystalin*". SFB, 2026

{c3} **Yasaman Kargar Gaz Kooh**, Yuwen Sun, Bahareh Bahmani, Ganesh Malayath, Leah Hayem, Debeni Devi Nongmaithem, Huanzhu Jiang, Guy Genin, Nathaniel Huebsch. "*Tunable Magnetorheological Elastomer Substrates for Studying Mechano-Genetic Interactions in Hypertrophic Cardiomyopathy*". SFB, 2026

{c4} **Yasaman Kargar Gaz Kooh**, Chen Zhao, Ganesh Malayath, Bahareh Bahmani, Leah Hayem, Ghiska Ramahdita, Huanzhu Jiang, Xiucui Ma, David Rawnsley, Abhinav Diwan, Nathaniel Huebsch. "*Mechanistic Characterization of CRYAB-R120G Desmin-Related Cardiomyopathy in Human iPSC-Derived 3D Micro-Heart Tissues*". ***Circulation Research***. BCVS, 2025

{c5} **Yasaman Kargar Gaz Kooh**, Ghiska Ramahdita, Ganesh Malayath, Harper Jiang, Xiucui Ma, Chen Zhao, David Rawnsley, Abhinav Diwan, Nathaniel Huebsch. "*Using Mechanical Cues to Study Desmin-related Cardiomyopathy in Human induced Pluripotent Stem Cells-derived Cardiac-micro Tissues*". SFB, 2025

{c6} Huanzhu Jiang, Ganesh Malayath, Ghiska Ramahdita, **Yasaman Kargar Gaz Kooh**, Sharon Cresci, Nathaniel Huebsch. "*Modeling Cardiomyocyte-fibroblast Crosstalk in Hypertrophic Cardiomyopathy Pathogenesis using iPSC-derived Micro Heart Tissue*". SFB, 2025

{c7} Ganesh Malayath, Huanzhu Jiang, **Yasaman Kargar Gaz Kooh**, Ghiska Ramahdita, Nathaniel Huebsch. "*Development of an Engineered Heart Tissue Platform to Study the Progressive Tissue Stiffness Change in Hypertrophic Cardiomyopathy*". SFB, 2025

PRESENTATION and TALKS

{p1} ***Yasaman Kargar Gaz Kooh***, "*Mechanical forces enable human stem cell modeling of protein aggregation cardiomyopathy*", Institute of Materials Science and Engineering, **Invited Oral talk**, 2026, Washington University McKelvey School of Engineering, St. Louis, Missouri, USA

{p2} ***Yasaman Kargar Gaz Kooh***, "*Low-dose JAK1 inhibitor partially rescues contractility in human micro-tissue model of CRYAB-linked Desmin-related cardiomyopathy*", DRSCB/Dev/CRM Retreat, Oral talk, 2026, Washington University School of Medicine, St. Louis, Missouri, USA

{p3} ***Yasaman Kargar Gaz Kooh***, "*Mechanical forces enable human stem cell modeling of protein aggregation cardiomyopathy*", Engineering in Cardiovascular Medicine, Poster, 2026, Ann Arbor, Michigan, USA **(Second Place, Best Poster Award)**

{p4} ***Yasaman Kargar Gaz Kooh***, "*A Micro-Engineered Heart Tissue Model of Desmin-related Cardiomyopathy Caused by Mutant B Crystallin*", Society for Biomaterials (SFB), Oral talk, 2026, Atlanta, Georgia, USA.

{p5} ***Yasaman Kargar Gaz Kooh***, "*Tunable Magnetorheological Elastomer Substrates for Studying Mechano-Genetic Interactions in Hypertrophic Cardiomyopathy*", Society for Biomaterials (SFB), Oral talk, 2026, Atlanta, Georgia, USA

{p6} ***Yasaman Kargar Gaz Kooh***, "*A Micro-Engineered Heart Tissue Model of Desmin-related Cardiomyopathy Caused by Mutant alpha-B Crystalin*", Center for Cardiovascular Research, Oral talk, 2025, Washington University School of Medicine, St. Louis, Missouri, USA

{p7} ***Yasaman Kargar Gaz Kooh***, "*Mechanical forces enable human stem cell modeling of protein aggregation cardiomyopathy*", Additional Ventures, Oral talk, 2025, Online

{p8} ***Yasaman Kargar Gaz Kooh***, "*Mechanistic Characterization of CRYAB-R120G Desmin-Related Cardiomyopathy in Human iPSC-Derived 3D Cardiac Micro-Tissues*", Center for Investigation of Membrane Excitability Diseases, Poster, 2025, Washington University School of Medicine, St. Louis, Missouri, USA

{p9} ***Yasaman Kargar Gaz Kooh***, "*Using Mechanical Cues to Study Desmin-related Cardiomyopathy in hiPSC-derived Cardiac Micro-Tissues*", Center for Cardiovascular Research, Oral talk, 2024, Washington University School of Medicine, St. Louis, Missouri, USA

{p10} ***Yasaman Kargar Gaz Kooh***, "*Dynamic Control of Contractile Resistance To iPSC-derived micro-heart muscle arrays*", Center for Cardiovascular Research, Oral talk, 2023, Washington University School of Medicine, St. Louis, Missouri, USA

TEACHING EXPERIENCES

- Teaching Assistant for Biomaterials Science course (BME523), Washington University in St. Louis **2024**

TECHNICAL SKILLS

- Proficient in various imaging microscopies: Epifluorescence Microscopy, Laser Scanning Confocal Microscopy
- Proficient in cell culture: induced Pluripotent Stem Cells, Cardiomyocytes, C2C12, Primary Cardiac Fibroblast
- Proficient in cardiac electrophysiology using Optogenetics (ChR2-P2A-RGECO1.2)
- Proficient in molecular and cellular biology: Molecular cloning (PCR, restriction digestion, ligation); Real-Time Polymerase Chain Reaction (qpcr), Western Blotting, Viral Transduction in Gene Therapy (AAV6), Flow Cytometry
- Proficient in software: AutoCAD, ProCast, SolidWorks, Autodesk 360

PROFESSIONAL SERVICES

Reviewer for Journal
Journal for Biomedical and Material Research (JMBR)

PROFESSIONAL Organizations

Society for Biomaterials (SFB): Graduate Student Member; 2024-present
American Heart Association (AHA): Graduate Student Member; 2024-present
International Society for Heart Research (ISHR): Graduate Student Member; 2024-2026

STUDENT SUPERVISION, Co-supervised with Prof. Nathaniel Huebsch

- Bahareh Bahmani (Ms Student)
- Javier Santiago Perez (MD/PhD Student)
- Yuwen Sun (Ms Student)
- Leah Hayem (Undergrad Student)
- Jenna Nguyen (Undergrad Student)
- Hertier Umuragwa (Undergrad Student)
- Joseph Harris (Undergrad Student)
- Alma Bendavid (Undergrad Student)
- Hayden Jingst (Undergrad Student)